

Incorrect Fault Indication

Symptoms

- Alarm lamp is **not** illuminated at the engine room panel or the remote panel when alarm condition is present.

How To Use This Tree

This symptom tree can be used to troubleshoot panel fault symptoms. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check Panel Alarm Lamp Indication	
	STEP 1A. Check Engine Room Panel Alarm Lamp Indication	Alarm lamp illuminated?
	STEP 1B. Check Remote Panel Alarm Lamp Indication	Alarm lamp illuminated?
STEP 2.	Check Engine Room Panel Wiring	
	STEP 2A. Check Engine Room Panel Alarm (Red Lamp) Supply Wire	Less than 10 ohms resistance?
	STEP 2B. Check Engine Room Panel Return Wire	Less than 10 ohms resistance?
STEP 3.	Check Panel System Cables	
	STEP 3A. Check Engine Room Panel Cable	Less than 10 ohms resistance?
	STEP 3B. Check Remote Panel Cable	Less than 10 ohms resistance?

STEPS		SPECIFICATIONS
STEP 4.	Check Customer Interface Box	
	STEP 4A. Check the Engine Room Panel Alarm (Red Lamp) Supply Wire	Less than 10 ohms resistance?
	STEP 4B. Check Remote Panel Alarm (Red Lamp) Supply Wire	Less than 10 ohms resistance?
	STEP 4C. Check Red Lamp Signal Wire	Less than 10 ohms resistance?
STEP 5.	Check Customer Interface Box to Engine Harness Cable	
	STEP 5A. Check Red Lamp Signal Wire	Less than 10 ohms resistance?
STEP 6.	Check Customer Interface Box Logic Unit	
	STEP 6A. Check Engine Room Panel Alarm (Red Lamp) Supply Terminal	24 VDC

STEP 1. Check Panel Alarm Lamp Indication

STEP 1A. Check Engine Room Panel Alarm Lamp Indication

Conditions :

- Locate engine room panel

Action	Specification/Repair	Next Step
Verify engine room panel alarm lamp is illuminated.	Alarm lamp illuminated? YES	1B
	Alarm lamp illuminated? NO	2A

STEP 1B. Check Remote Panel Alarm Lamp Indication

Conditions :

- Locate remote panel

Action	Specification/Repair	Next Step
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Verify remote panel alarm lamp is illuminated.	Alarm lamp illuminated? YES	Repair complete.
	Alarm lamp illuminated? NO	3B

STEP 2. Check Engine Room Panel Wiring

STEP 2A. Check Engine Room Panel Alarm (Red Lamp) Supply Wire

Conditions :

- Locate engine room panel
- Open engine room panel door.

Action	Specification/Repair	Next Step
Check wires between harness connector and control panel connector. • Disconnect cable C14 from the engine room panel • Place one test lead on the engine room panel alarm (red lamp) supply pin on connector C14. • Place the other test lead on the engine room panel alarm (red lamp) supply pin on the control panel connector.	Less than 10 ohms resistance? YES	2B
	Less than 10 ohms resistance? NO Repair : Replace the faulty wire. Refer to Procedure 015-024 (/qs3/pubsys2/xml/en/procedures/115/115-015-024.html).	Repair complete.

STEP 2B. Check Engine Room Panel Return Wire

Conditions :

- Locate engine room panel
- Open engine room panel door.

Action	Specification/Repair	Next Step
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Check wires between harness connector and control panel connector. • Disconnect cable C14 from the engine room panel. • Place one test lead on the engine room panel return pin on connector C14. • Place the other test lead on the engine room panel return pin on the control panel connector.	Less than 10 ohms resistance? YES	3A
	Less than 10 ohms resistance? NO Repair : Replace the faulty wire. Refer to Procedure 015-024 (/qs3/pubsys2/xml/en/procedures/115/115-015-024.html).	Repair complete.

STEP 3. Check Panel System Cables

STEP 3A. Check Engine Room Panel Cable

Conditions :

- Disconnect cable connector C14 from the engine room panel
- Disconnect cable connector C7 from the customer interface box.

Action	Specification/Repair	Next Step

Check the engine room panel cable. • Install a jumper between engine room panel alarm (red lamp) supply pin and the engine room panel return pin in connector C14. • Place the other test lead in the engine room panel alarm (red lamp) supply pin in connector C7. • Place the other test lead in the engine room panel return pin in connector C7.	Less than 10 ohms resistance? YES	4A
	Less than 10 ohms resistance? NO Repair : Replace the cable.	Repair complete.

STEP 3B. Check Remote Panel Cable		
Conditions : • Locate and open customer interface box • Locate and open remote panel • Disconnect remote panel cable from customer interface box X4 connector.		
Action	Specification/Repair	Next Step
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Check the remote panel cable. • Install a jumper between remote panel alarm (red lamp) supply terminal and the remote panel return terminal on the remote control panel X4 in the remote control panel. • Place one test lead on the remote panel alarm (red lamp) supply wire. • Place the other test lead on the remote panel return wire.	Less than 10 ohms resistance? YES	4B
	Less than 10 ohms resistance? NO Repair : Replace the cable.	Repair complete.

STEP 4. Check Customer Interface Box

STEP 4A. Check the Engine Room Panel Alarm (Red Lamp) Supply Wire

Conditions :

- Open the customer interface box
- Disconnect engine room cable at connector C7 from the customer interface box.

Action	Specification/Repair	Next Step

Check the engine room panel alarm (red lamp) supply wire. - Place one test lead on the engine room panel alarm (red lamp) supply pin in connector C7. - Place the other test lead on the engine room panel alarm (red lamp) supply terminal on the customer interface box logic unit.	Less than 10 ohms resistance? YES	4C
	Less than 10 ohms resistance? NO Repair : Replace the faulty wire. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/115/115-015-023.html).	Repair complete.

STEP 4B. Check Remote Panel Alarm (Red Lamp) Supply Wire

Conditions : Open the customer interface box		
Action	Specification/Repair	Next Step
Check the remote panel alarm (red lamp) supply wire. - Place one test lead on the remote panel alarm (red lamp) supply wire terminal in customer interface box X4 connector. - Place the other test lead on the remote panel alarm (red lamp) supply terminal of the customer interface box logic unit.	Less than 10 ohms resistance? YES	4C
	Less than 10 ohms resistance? NO Repair : Replace the faulty wire. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/115/115-015-023.html).	Repair complete.

STEP 4C. Check Red Lamp Signal Wire

Conditions : - Open the customer interface box - Disconnect the customer interface box to engine harness cable at the C3 connector.

Action	Specification/Repair	Next Step
Check the red lamp signal wire. - Place one test lead on the red lamp signal pin in connector C3. - Place the other test lead on the red lamp signal terminal of the customer interface box logic unit.	Less than 10 ohms resistance? YES	5A
	Less than 10 ohms resistance? NO Repair : Replace the faulty wire. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/115/115-015-023.html).	Repair complete.

STEP 5. Check Customer Interface Box to Engine Harness Cable

STEP 5A. Check Red Lamp Signal Wire

Conditions :

- Disconnect cable connector C10 from the engine harness
- Disconnect cable connector C3 from the customer interface box.

Action	Specification/Repair	Next Step
Check the customer interface box to engine harness cable. - Place one test lead in the red lamp signal pin in connector C10. - Place the other test lead in the red lamp signal pin in connector C3.	Less than 10 ohms resistance? YES	6A
	Less than 10 ohms resistance? NO Repair : Replace the cable.	Repair complete.

STEP 6. Check Customer Interface Box Logic Unit

STEP 6A. Check Engine Room Panel Alarm (Red Lamp) Supply Terminal

Conditions :

- Locate customer interface box
- Open the customer interface box.

Action	Specification/Repair	Next Step
Verify the engine room panel alarm (red lamp) supply terminal of the customer interface box logic unit is 24 VDC.	24 VDC YES Repair : Replace the faulty control panel. Refer to Procedure 015-024 (/qs3/pubsys2/xml/en/procedures/115/115-015-024.html) or 015-025 (/qs3/pubsys2/xml/en/procedures/115/115-015-025.html).	Repair complete.
	24 VDC NO Repair : Replace customer interface box logic unit after verifying on-engine harness and engine control module are operating property. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/115/115-015-023.html).	Repair complete.

